Controlled Source Seismics metadata documentation

Version: 0.3
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Scope

The Controlled Source Seismics (CSS) metadata documentation helps the end user to understand the CSS metadata model and its implementation as published CSS metadata catalogue. It shortly touches the realisation of the latter as a geodatabase and the related CSS discovery data service, but it does not provide comprehensive documentation for the service usage (which is available in the online documentation).

The Controlled Source Seismics metadata model

This is the initial publication of the CSS metadata model with example data only, as the basis for establishing the CSS metadata catalogue. Following publications will center around the metadata catalogue while always providing the latest metadata model and supplementary information/files.

The CSS metadata model was developed for transnational/organisational data integration of CSS data. Its intention is to provide as detailed scientific information about the described data sets as possible while restricting the obligatory "core" information to attributes that are commonly available in existing national and organisational contexts. Thus, the metadata model is a compromise that on the one hand promotes rich scientific metadata (discovery data) that allow a scientist to decide whether a data set is fit for the intended purpose or not. On the other hand, existing data providers are not excluded due to an incompatibility of their existing metadata models with the minimum requirements of the CSS metadata model. For details on the model development, please refer to DOI (UPDATE WHEN PAPER DOI IS AVAILABLE).

The attributes of the CSS metadata model are described below. The present publication includes the implementation of the metadata model in GeoJSON, which is supplied as the JSON-Schema document *"epos-css-schema.json"*. The related controlled vocabularies are provided as SKOS Turtle files. Proper publication and registration of the vocabularies is planned for the first full release of the CSS metadata catalogue. Sample metadata are provided for single and multiple data sets.

Core attributes (obligatory)

surveyName [CharacterString]:
Multiplicity: ([1], Allow duplicates: 0, Is ordered: False)
Name of the individual survey.

campaignName [CharacterString]:
Multiplicity: ([1], Allow duplicates: 0, Is ordered: False)
Name of seismic campaign/project.

acquisitionYear [Integer]:
Multiplicity: ([1], Allow duplicates: 0, Is ordered: False)
The year in which the data acquisition was conducted.

surveyType [SurveyTypeCode]:
Multiplicity: ([1], Allow duplicates: 0, Is ordered: False)
The geometry of the acquisition.
Vocabulary: survey_type.ttl

investigationMethod [InvestigationTypeCode]:
Multiplicity: ([1], Allow duplicates: 0, Is ordered: False)
The principle of seismic investigation.
Vocabulary: investigation_method.ttl

content [ContentCode]:
Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False)
The content of the dataset.
Vocabulary: processing_level.ttl

domain [SurveyDomainCode]:
Multiplicity: ([1], Allow duplicates: 0, Is ordered: False)
The domain (land, marine) in which the survey was conducted.
Vocabulary: domain.ttl

license [LicenseCode]:
Multiplicity: ([1], Allow duplicates: 0, Is ordered: False)
The license that applies to the dataset (that is described by this instance of metadata).
Vocabulary: license.ttl

uid and/or **dataUri**
Multiplicity: ([1], Allow duplicates: 0, Is ordered: False):

> **uid** [URI]:
> ⇒ *"identifier"*
> Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False), ([1] if **dataUri** is not provided)
> The URI that identifies and points to the landing page of the data publication/data set (a DOI, if it exists).
>
> **dataUri** [URI]:
> Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False), ([1] if **uid** is not provided)
> An URI that points directly to the data set (file). If several data sets exist, it should point to the highest level data set.

provenance [ProvenanceCode]:
Multiplicity: ([1], Allow duplicates: 0, Is ordered: False)
Provides a hint on the origin of the available digital data, e.g. "paper to digital", "analogue to digital", "digital", or similar.
Vocabulary: provenance.ttl

geolocated [Boolean]:
Multiplicity: ([1], Allow duplicates: 0, Is ordered: False)
Is the dataset geolocated? For raw data: source and receiver positions, for processed data coordinates of the data bins.

geometry [Geometry]:
Multiplicity: ([1], Allow duplicates: 0, Is ordered: False)
A spatial geometry (polygon, line, point, multiline) that describes the surface coverage of the survey.

size [Real]:
Multiplicity: ([1], Allow duplicates: 0, Is ordered: False)
The size of the survey. "length [km]" if "survey geometry" is "2D", "area [km2]" if "survey geometry" is "3D". This attribute is computed from the geometry.

Descriptive attributes (optional)

clientName [CharacterString]:
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)
The organisation that initiated the survey (project owner).

clientId [URI]:
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)
Preferably ROR, or other resolvable identifier

contractorName [CharacterString]:
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)
The organisation that was responsible for conducting the survey.

contractorId [URI]:
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)
Preferably ROR, or other resolvable identifier

purpose [SeismicSurveyPurposeCode]:
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)
The original purpose for which the survey was conducted.
Vocabulary: purpose.ttl

sourceType [SeismicSourceTypeCode]:
Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False)
The type of the seismic source.
Vocabulary: source_type.ttl

sourceParameters [CharacterString]:
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)
Information on e.g. sweep or charge size.

sensorType [SeismicSensorTypeCode]:
Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False)
The type of the seismic sensor.
Vocabulary: sensor_type.ttl

sensorParameters [CharacterString]:
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)
Information on e.g. model, frequency response.

verticalScaleUnit [UnitCode]:
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)
The unit of the vertical scale. Required if "vertical scale" is provided.
Vocabulary: vertical_scale_unit.ttl

verticalScale [Real]:
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)
The value of the vertical scale. Required if "vertical scale unit" is provided.

Technical attributes (optional)

spread [SeismicSpreadTypeCode]:
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)
The type of the prevailing spread used during the survey.
Vocabulary: spread.ttl

processing [SeismicProcessingLevelCode]:
Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False)
A concise description of the processing steps that led to the described dataset.
Vocabulary: processing_level.ttl

description [CharacterString]:
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)
Text with additional information on the survey purpose and/or summary of results, e.g. abstract.

detailedDocumentation [URI]:
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)
Report, publication or similar with detailed information on the survey.

velocityModel [Boolean]:
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)
Is a velocity model available/specified?

preview [URI]:
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)
A preview image for the dataset.

spacingSource [Real]:
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)
Prevailing distance between source points [m].

spacingReceiver [Real]:
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)
Prevailing distance between receiver points [m].

samplingRate [Real]:
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)
The sampling rate used while recording [ms].

countSourcepoints [Integer]:
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)
Total number of source points.

countReceiverpoints [Integer]:
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)
Total number of receiver points.

offsetNear [Real]:
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)
Minimum offset between source and receiver [m].

offsetFar [Real]:
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)
Maximum offset between source and receiver [m].

countChannels [Integer]:
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)
Number of channels of the acquisition system

foldNominal [Integer]:
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)
The nominal fold of survey.

specification [CharacterString]:
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)
Text with additional information on the survey design & acquisition
(including environmental conditions and other factors that could have an influence on the data).

Data attributes

format [DataFormatCode]:
Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False)
Format of the data file.
Vocabulary: format.ttl

volume [Real]:
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)
Size of the data file [GB].

legalConstraints [CharacterString]:
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)
A description of legal constraints, in particular for datasets without license or with restricted access.

datePublication [Date]:
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)
Date of first availability or publication.

dateMoratorium [Date]:
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)
End date of a moratorium/embargo period. When this attribute is filled, the license and legal information should describe the status after the end of the moratorium/embargo.

Data provision attributes

dataProviderName [CharacterString]:
Multiplicity: ([1], Allow duplicates: 0, Is ordered: False)
The organisation that is responsible for the instance of discovery data.

dataProviderId [URI]:
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)
Preferably ROR, or other resolvable identifier

contactName [CharacterString]:
Multiplicity: ([1], Allow duplicates: 0, Is ordered: False)
Name of contact person, possibly with contact information if not available as URI.

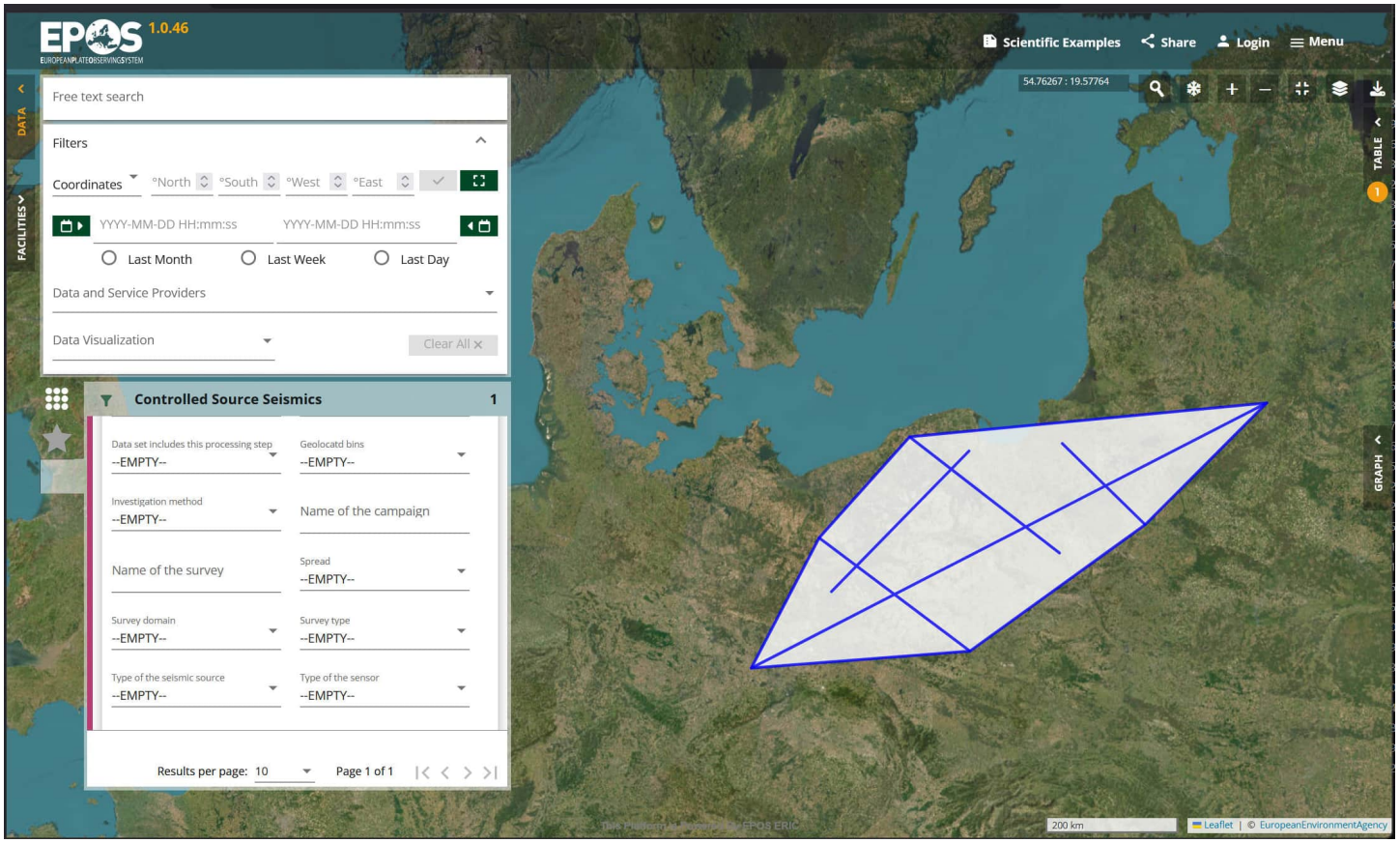
contactId [URI]:
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)
Preferably ORCID, or other resolvable identifier

contactURI [URI]:
Multiplicity: ([1], Allow duplicates: 0, Is ordered: False)
Link to contact information.

dateUpdated [date]:
Multiplicity: ([1], Allow duplicates: 0, Is ordered: False)
Date when the record was last updated.

The CSS discovery data service

The CSS metadata model is implemented for EPOS as a PostGIS geodatabase that contains the latest release of the CSS metadata catalogue. The CSS discovery service is a Web Feature Service (WFS) that follows the specifications of the Open Geospatial Consortium (OGC) standard WFS 2.0.2, which is related to ISO 19119 standard Geographic information - Services. This service was tailored for providing discovery functionality for CSS data sets to the EPOS Platform, but it can also be interfaced directly. For details, please refer to the online documentation.



Screenshot of the CSS discovery service on a development instance of the EPOS Platform.

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